

$$\int_0^{\frac{\pi}{6}} \frac{\sqrt{3}\sin t}{\cos^2 t}\,dt$$

Let $u = \cos t$ $\quad \frac{du}{dt} = -\sin t \rightarrow dt = -\frac{du}{\sin t}$

when $t = \frac{\pi}{6}$, $u_1 = \cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}$

$t = 0$, $u_2 = \cos 0 = 1$

$$\int_1^{\frac{\sqrt{3}}{2}} \frac{\sin t}{u^2} \cdot \left(-\frac{1}{\sin t}\right) du$$

$$= -\int_1^{\frac{\sqrt{3}}{2}} \frac{1}{u^2}\,du$$

$$= \left[u^{-1}\right]_1^{\frac{\sqrt{3}}{2}}$$

$$= \left[\frac{1}{\cos t}\right]_0^{\frac{\sqrt{3}}{2}}$$ 不用按回去，范围已经改了

$$= \left[\frac{1}{\cos\frac{\sqrt{3}}{2}} - \frac{1}{\cos 1}\right] = -0.30$$

$$\left(\frac{\sqrt{3}}{2}\right)^{-1} - 1 = 0.154$$

$$\int_1^2 \frac{e^{\frac{1}{x}}}{x^2}\,dx$$

Let $u = e^{\frac{1}{x}}$ $\quad \frac{du}{dx} = e^{\frac{1}{x}} \cdot (-x^{-2}) = -\frac{u}{x^2}$

$$dx = -\frac{x^2}{u} \cdot du$$

when $x = 2$, $u_1 = e^{\frac{1}{2}}$

$x = 1$, $u_2 = e^{1}$

$$\int_1^2 -\,dx = \int_{e}^{e^{\frac{1}{2}}} \frac{u}{x^2} \cdot \left(-\frac{x^2}{u}\right) du$$

$$= -\int_{e}^{e^{\frac{1}{2}}} \cdot 1\, du$$

$$= -\left[u\right]_{e}^{e^{\frac{1}{2}}}$$

$$= -e^{\frac{1}{2}} + e^{1} = 1.0696$$

vertical integration:

$\sqrt[3]{x} = \frac{1}{x}$ $\qquad x^{\frac{1}{3}} \qquad x^{\frac{4}{3}} \qquad (x^{-1})$

$x = 1$

$$\int_1^8 \sqrt[3]{x} - \frac{1}{x}\, dx = \left[\frac{3}{4} \cdot x^{\frac{4}{3}} - \ln|x|\right]_1^8$$

$$= \left[\left(\frac{3}{4} \cdot 8^{\frac{4}{3}} - \ln|8|\right) - \left(\frac{3}{4} - \ln|1|\right)\right]$$

$$= 9.706$$

Irrational integration

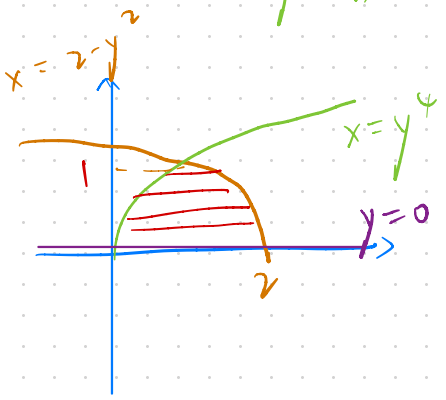
$$\int_{-1}^{1} e^y - y^2 + 2 \, dy = \left[e^y - \frac{y^3}{3} + 2y \right]_{-1}^{1}$$

$$= e^1 - \frac{1}{3} + 2 - \left(e^{-1} + \frac{1}{3} - 2 \right)$$

$$= 1.683$$

$x = y^4$ 、 $y = \sqrt{2-x}$ → 交於 $x \le 2$

↳ $y = \sqrt[4]{x}$ $(x \ge 0)$



vertical integration

$\sqrt{2-x} = \sqrt[4]{x}$

$(2-x)^2 = x$

$x \quad -4$
$x \quad -1$

$4 - 4x + x^2 - x = 0$

$x^2 - 5x + 4 = 0$

$(x-4)(x-1) = 0$

$x = 4$ or $x = 1$ 代入 $x = y^4 = 1$

(rej)

horizontal

$2 - x = y^2$

$x = 2 - y^2$

$\int_0^1 2 - y^2 - y^4 \, dy$

$= \left[2y - \frac{y^3}{3} - \frac{y^5}{5}\right]_0^1$

$= 2 \times 2 - \frac{2^3}{3} - \frac{2^5}{5}$ $\quad 2 - \frac{1}{3} - \frac{1}{5}$

$= -5.067$ $\quad = \frac{22}{15}$

$2 - y^2 = y^4$

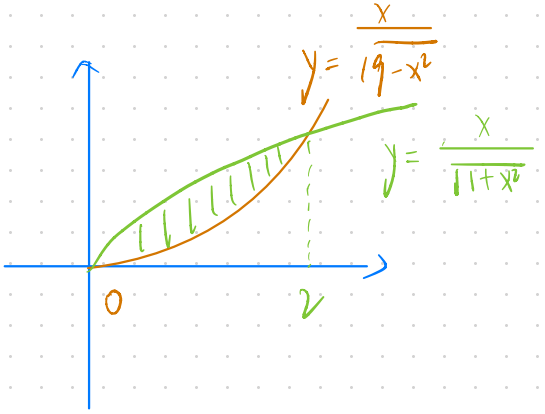
$-y^4 - y^2 + 2 = 0$

$(y^2+2)(y^2-1) = 0$

$y^2 = -2$ (rej)

$y = +1$ or $y = -1$ (rej)

$$y = \frac{x}{\sqrt{1+x^2}} \quad ; \quad y = \frac{x}{\sqrt{9-x^2}} \quad , \quad x \geq 0$$



$$\frac{x}{\sqrt{1+x^2}} = \frac{x}{\sqrt{9-x^2}}$$

$$\frac{x^2}{1+x^2} = \frac{x^2}{9-x^2}$$

$$x^2(9-x^2) = x^2(1+x^2)$$

$$9x^2 - x^4 - x^2 - x^4 = 0$$

$$-2x^4 + 8x^2 = 0$$

$$x^2(-2x^2+8) = 0$$

$$x = 0 \quad \text{or} \quad x = 2$$

$$\int_0^2 \frac{x}{\sqrt{1+x^2}} - \frac{x}{\sqrt{9-x^2}}\, dx$$

$$= \int_0^2 \frac{x}{\sqrt{1+x^2}}\, dx - \int_0^2 \frac{x}{\sqrt{9-x^2}}\, dx$$

let $u_1 = 1+x^2$, $du_1 = 2x\,dx \rightarrow dx = \frac{du_1}{2x}$

when $x = 2$, $u_1 = 1+4 = 5$

$x = 0$, $u_1 = 1$

let $u_2 = 9-x^2$, $du_2 = -2x\,dx \rightarrow dx = -\frac{du_2}{2x}$

when $x = 2$, $u_2 = 9-4 = 5$

$x = 0$, $u_2 = 9-0 = 9$

$$= \int_0^2 \frac{x}{\sqrt{u_1}} \cdot \frac{du_1}{2x} - \int_9^5 \frac{x}{\sqrt{u_2}} \cdot \left(-\frac{1}{2x}\right) du_2$$

$$= \frac{1}{2}\int_1^5 \frac{1}{\sqrt{u_1}}\, du_1 + \frac{1}{2}\int_9^5 \frac{1}{\sqrt{u_2}}\, du_2 = \left[u_1^{\frac{1}{2}}\right]_1^5 + \left[u_2^{\frac{1}{2}}\right]_9^5 = 0.4721$$

$$= \left[u_1^{\frac{1}{2}}\right]_1^5 - \left[u_2^{\frac{1}{2}}\right]_9^5 = -4(\sqrt{5}-0) - 2(3-\sqrt{5}) =$$